



Original article

Is Delayed Driving Licensure Associated With Emerging Adult Health, Education, and Employment?



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A B S T R A C T

Purpose: Driving licensure remains a major developmental milestone for adolescents as they become more independent to access important health, education, and employment opportunities. Today, more teens are delaying driving licensure than before. We investigated associations of delayed licensure with health, education, and employment 4 years after high school.

Methods: We analyzed data from all seven annual assessments (W1–W7) of the NEXT Generation Health Study, a nationally representative cohort survey starting at 10th grade (W1, 2009–2010). The independent variable was delaying driving licensure (DDL [delaying ≥ 1 year] vs. No-DDL), defined as participants receiving driver licensure ≥ 1 year after the initial legal eligibility time until W7. Outcome variables were self-reported health, education, and employment at W7. Covariates included sex, race/ethnicity, family affluence, parental education, and urbanicity. Multinomial logistic regressions were conducted considering complex survey features.

Results: No-DDL versus DDL was associated with a higher likelihood of (1) excellent (adjusted odds ratio [AOR] = 2.06, $p < .001$), good (AOR = 1.74, $p < .001$), and fair (AOR = 1.34, $p = .008$) health compared with poor health; (2) completing a 4-year college or graduate school [AOR = 2.71, $p < .001$] and tech/community college [AOR = 1.92, $p = .004$] compared with high school or less; and (3) working ≥ 30 hours/week (AOR = 7.63, $p = .011$) and working < 30 hours/week (AOR = 1.54, $p = .016$) compared with not working.

Discussion: Among emerging adults, no delay in driving licensure was associated with better self-reported health, higher education, and more working hours four years after leaving high school. Although earlier driving licensure increases driving exposure and risk, avoiding DDL appears to provide advantages for health, education, and employment during early adulthood.

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IMPLICATIONS AND CONTRIBUTIONS STATEMENT

Although teen licensure is viewed as a major developmental milestone, nearly 70% of eligible adolescents delayed licensure at least 1 year. Those without licensure delay reported better health, education, and employment in emerging adulthood. Future study should assess DDL contextual facets and provide details of measured early adulthood success.

Conflicts of interest: All authors have no conflicts of interest to be declared.

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Transportation is a vital structural facet of the day-to-day workings in society, communities, and the lives of people [1]. At the individual level, the word transportation typically calls up an image of someone using public transportation and, even more

commonly, someone driving a car or truck [1]. Reliable personal transportation provides people valuable independence. Specifically, for emerging adults, the greater independence as a result of driving could facilitate a broad array of opportunities such as attending health care–related visits, participating in supplemental education opportunities (e.g., volunteer positions to gain educational experience), and/or taking an off-campus part-time or full-time employment [2,3].

Although teen licensure is viewed as a major developmental and transitional milestone, studies consistently show that learning to drive and becoming licensed are accompanied by measurable risk of harm and injury due to crashes. Adolescents in the United States are at high risk of nonfatal injury, physical disability, and death caused by motor vehicle crashes [4]. In 2018, 1,719 young drivers were killed in motor vehicle crashes, and 199,000 were injured [5]. Motor vehicle crashes remain the leading cause of death for teens and young adults through age 24 years [6].

To prevent and reduce fatal and nonfatal injuries among novice teen drivers, graduated driver licensing (GDL) programs were developed to introduce new young drivers to safely gain driving experience before obtaining full independent driving privileges [7]. GDL programs reduce teen crashes and deaths. Comprehensive GDL programs have contributed to the reduction fatal crashes among teen drivers in the United States by as much as 20%–40% [8]. In the United States, all states have a GDL program that includes three stages: learner's permit, intermediate license, and unrestricted license, although the content (i.e., types of restrictions) of these stages varies across states [9].

Overall, becoming a licensed driver through the safety benefits offered as a result of GDL and having reliable independent transportation may considerably enhance short- and longer-term well-being/health, education, and employment for young drivers [2,3,10,11]. Despite the potential benefits of obtaining licensure as a teen novice driver, there has been a downward trend of obtaining licensures among teens. One epidemiological study has shown that the rate of obtaining unsupervised driving licenses among US high school students has dropped from 81% to 72% during 2006–2015 [12]. More recently, a nationally representative US young cohort study has shown that delaying driver licensure was widespread, and nearly 70% of eligible adolescents delayed at least one year to obtain their licensure during 2010–2017 [13].

Accordingly, if teens delay their licensure and then later pursue licensure outside the age parameters of their state GDL policy, it is conceivable that they may miss out on graduated driving safety practices that are intentionally tailored to bolster teen driver safety and lower crash risk among young novice drivers [13,14]. Although delaying driving licensure (DDL) may contribute to lower teens' crash risk owing to no exposure to driving [15], both crash risk and crash rates are still higher later when older teens/emerging adults begin to drive without much, if any, prior novice driver safety instructions and/or supervised driving experience [16,17]. Thus, it is reasonable to suggest that many drivers (i.e., older teens ≥ 18 y/o and emerging adults) who have delayed licensure may initially have a heightened risk for crash. Such little-to-no prior driving exposure, experience, and instruction among older teens/emerging adults who pursue licensure beyond GDL policy age limits could conceivably compromise some of their own subsequent beneficial opportunities related to health, education, and employment [18,19]. In addition to potentially experiencing increased crash-related injuries and disabilities, DDL may contribute to other important consequences (e.g., getting traffic tickets, license suspension) resulting from increased risky driving

behaviors (e.g., speeding, distracted driving, tailgating) among older teens/emerging adults owing to the unique developmental considerations (e.g., sensation seeking, lower inhibitory control, less efficient working memory, or poorer attention) [20]. These consequences may impact teens' future health, education, and employment opportunities to different extents [21].

Taken together, there is a need to further understand measurable effects that DDL might have on health, education, and employment among emerging adults. The purpose of this study was to evaluate the associations of DDL among adolescents with their health, education, and employment 4 years after high school.

Methods

Sampling

We used the data from all seven waves (W1–7) of the NEXT Generation Health Study that encompassed a nationally representative youth cohort starting at W1 in 10th grade (2009–2010) and ending four years after leaving high school. The NEXT study used a multistage sampling strategy to select primary sampling units (US school districts in the NEXT study) from nine census divisions. Specifically, from each school district, schools and classrooms within the schools were randomly selected. Among the 145 selected schools, 81 schools agreed to participate. African-Americans were oversampled to obtain a large enough sample ($N = 687$) for more accurate population estimates. Surveys were administered annually for each wave in the spring semester. In total, there were 2,783 participants who enrolled in the NEXT study. From W1 to W7, 90% ($N = 2,524$), 87% ($N = 2,439$), 86% ($N = 2,395$), 78% ($N = 2,177$), 79% ($N = 2,202$), 83% ($N = 2,306$), and 83% ($N = 2,323$) of the 2,783 participants completed the survey each year, respectively. Parents provided informed consent for their child's participation. Participants younger than 18 years of age provided assent, and consent was obtained from participants once they reached 18 years of age. The Institutional Review Board of the Eunice Kennedy Shriver National Institute of Child Health and Human Development reviewed and approved the study protocol.

Independent variable

Delaying driving licensure (DDL, W1–7). DDL was derived using participants' age at the time of the survey, information on their state's legal requirement, and their self-reported driving licensure status. Participant date of birth was assessed at baseline, and their age at each wave was calculated. State requirements for initial legal eligibility for getting an independent driver's license were obtained for NEXT survey years through Department of Motor Vehicles websites for corresponding states. Driving licensure was ascertained by asking, "Do you have a driver's license?", with four possible responses: (1) no license of any sort, (2) permit to take the classroom component of driver education only, (3) permit allowing supervised practice driving with an instructor or licensed adult, and (4) license allowing independent, unsupervised driving (with or without temporary restrictions on late night driving, teen passengers, etc.) from all seven waves. Therefore, those participants who did not report licensure status at any wave were set to missing in the analyses. We defined DDL as a dichotomous variable, i.e., No-DDL versus DDL, with those in the delayed group having not obtained their independent license (i.e., response 4) at the first wave in which they were eligible.

Table 1

Participant characteristics and outcome measures stratified by delaying driving licensure status

	No-DDL		DDL (≥ 1 year)		Total		Rao-Scott χ^2	p
	N	%	N	%	N	%		
Overall ^a	560	30.71	1,965	69.29	2,525	100	16.208	<.001
Sample characteristics								
Race ^b								
Latinos	38	10.03	659	89.97	697	100	35.988	<.001
Non-Latino Blacks	43	18.04	392	81.96	435	100		
Non-Latino Whites	438	42.33	601	57.67	1,039	100		
Others ^d	23	38.71	92	62.29	115	100		
Sex ^b								
Male	248	32.50	761	67.50	1,009	100	.213	.645
Female	294	31.08	986	68.92	1,280	100		
Parent education ^b								
High school or less	105	19.51	716	80.49	821	100	29.226	<.001
Some college	224	34.01	584	65.99	808	100		
Bachelor or more	194	42.69	333	57.31	527	100		
Urbanicity ^b								
Urban	61	17.04	729	82.96	790	100	4.080	.130
Suburban	164	27.70	612	72.30	776	100		
Rural	317	43.12	406	56.88	723	100		
Family affluence ^b								
Low	57	13.66	633	86.34	690	100	46.797	<.001
Moderate	274	31.90	813	68.10	1,087	100		
High	211	45.51	301	54.49	512	100		
W1 health ^b								
Excellent	121	40.36	269	59.64	390	100	37.819	<.001
Good	299	33.88	855	66.12	1,154	100		
Fair	104	23.04	515	76.96	619	100		
Poor	16	18.39	100	81.61	116	100		
W7 health ^c								
Excellent	96	39.21	254	60.79	350	100	15.412	.002
Good	286	33.09	913	66.91	1,199	100		
Fair	104	24.07	504	75.93	608	100		
Poor	5	11.48	113	88.52	118	100		
W7 education ^c								
High school or less	81	18.13	616	81.87	697	100	45.571	<.001
Tech/community college	211	31.64	813	68.36	1,024	100		
4-year college+	195	45.23	351	54.77	546	100		
W7 work status (working hrs./week) ^c								
None	279	27.59	1,055	72.41	1,334	100	5.970	.051
<30 hours	199	35.74	688	64.26	897	100		
≥ 30 hours	9	55.67	22	44.33	31	100		

DDL = delaying driving licensure.

^a Weighted % using the overall weight based on W1–W7 weights.^b Weighted % using W1 weight.^c Weighted % using W7 weight.^d Including Asian, American, Indian or Alaska Native, and Native Hawaiian or other Pacific Islanders.

Outcome variables

Overall physical health in emerging adulthood (W7). Perceived overall physical health was measured with a single question, “Would you say your health is...?” with four possible responses, excellent, good, fair, and poor at W7 [22].

Education attainment (W7). Education attainment was measured with one question, “What is the highest grade of regular school you have completed (or anticipate completing by the end of the current academic term)?” with seven possible responses: less than high school diploma, high school diploma, General Educational Diploma (GED), some college or technical school, associate’s degree, bachelor’s degree, and graduate degree. The seven categories were collapsed into three categories: less than or equal to high school diploma or GED; some college, technical school, or associate’s degree; and bachelor’s degree or graduate degree.

Employment in emerging adulthood (W7). Employment and volunteer work were measured with one question, “On average, what are the total hours per week you spend working in paid and/or unpaid jobs?” with possible responses: none, ≤ 5 , 6–10, 11–15, 16–20, 21–25, 26–30, and > 30 hours/week. The eight categories were collapsed into three categories: none, < 30 , and ≥ 30 hours/week.

Demographics and covariates

Participants’ sex, race/ethnicity, parental education level, family affluence, overall physical health, and urbanicity were collected at the baseline visit (W1, 10th grade high school). The biological sex of participants was female or male. Participant race/ethnicity was categorized as Latinos, non-Latino Blacks, non-Latino Whites, and other (including Asians, American Indians or Alaska Natives, and Native Hawaiians or other Pacific Islanders). The highest education attainment of participants’

Table 2

Adjusted multinomial logistic regressions of overall physical health at W7 on delaying driving licensure

	Excellent			Good			Fair		
	AOR	95% CI	p	AOR	95% CI	p	AOR	95% CI	p
DDL (≥ 1 yr.)	Ref.	–	–	Ref.	–	–	Ref.	–	–
No-DDL	1.67	2.19, 7.12	<.001	1.45	1.81, 5.61	<.001	1.21	1.29, 4.98	.012

Note: Logit models used poor as the reference category.

AOR = adjusted odds ratio; CI = confidence interval; DDL = delaying driving licensure.

^aadjusting for sex, race/ethnicity, family affluence, parental education, urbanicity, and overall physical health at W1.

parents was defined as the following: (1) less than a high school diploma, a high school diploma, or GED; (2) some college, technical school, or associate degree; or (3) bachelor or graduate degree. The family socioeconomic status was obtained by the Family Affluence Scale, a widely used and validated proxy for socioeconomic status [23]. Specifically, participants were asked how many cars their family owned, the number of computers owned, whether participants had their own bedroom, and the number of family vacations taken in the last 12 months. Then, participants' family affluence status was categorized as low, moderate, or high affluence [24]. Adolescent-perceived overall physical health was measured with a single question, "Would you say your health is...?" with four possible responses, excellent, good, fair, and poor at W1 [22]. Urbanicity (suburban, rural, vs. urban) was derived from participants' school location at W1 according to the National Center for Education Statistics [25].

Statistical analyses

Multinomial logistic regressions were conducted to assess associations of DDL among adolescents with their health, education, and employment 4 years after high school. Unadjusted and adjusted model analyses were conducted. The adjusted models were controlled for sex, race/ethnicity, family affluence, urbanicity, parental education attainment, and W1 general health. All statistical analyses were performed using SAS software, version 9.4, (SAS Institute, Cary, NC) taking complex survey sampling features (i.e., strata [i.e., the nine census divisions], clustering [i.e., primary sampling unit], and sampling weights) into account [26]. Odds ratio (OR), adjusted odds ratio (AOR), and 95% confidence interval (95% CI) were reported. The statistical significance level was set at $p = .05$ for the analyses.

Results

Of 2,783 participants, 258 were excluded from the analyses because they did not report licensure status at any wave. Table A1 shows demographic characteristics of included and excluded participants. The weighted mean ages of participants at W1, W2, W3, W4, W5, W6, and W7 were 16.27 years (standard

error [SE] = 0.03), 17.19 years (SE = 0.03), 18.17 years (SE = 0.03), 19.16 years (SE = 0.02), 20.28 years (SE = 0.02), 21.28 years (SE = 0.02), and 22.64 years (SE = 0.03), respectively. Table 1 shows participant characteristics and outcome measures stratified by delaying licensure status. Of 2,525 participants who were legally eligible for licensure through W7, 1,965 (69.29%, weighted and hereafter) reported DDL. The results of the Rao-Scott χ^2 test show that DDL is significantly associated with race/ethnicity ($\chi^2 = 41.65, p < .001$), parent education ($\chi^2 = 27.11, p < .001$), and family affluence ($\chi^2 = 58.38, p < .001$). In addition, the results of the Rao-Scott χ^2 test show that DDL is significantly associated with W1 overall physical health ($\chi^2 = 37.82, p < .001$), W7 overall physical health ($\chi^2 = 15.41, p = .002$), education attainment ($\chi^2 = 45.57, p < .001$), and employment indicated by working hours per week ($\chi^2 = 5.97, p = .050$).

Table 2 shows the associations between DDL and overall physical health 4 years after high school from adjusted multinomial logistic regression models. Compared with DDL, no-DDL was associated with a higher likelihood of excellent (AOR = 1.67, 95% CI = 2.19, 7.12), good (AOR = 1.45, 95% CI = 1.81, 5.61), and fair (AOR = 1.21, 95% CI = 1.29, 4.98) health compared with poor health.

Table 3 shows the associations between DDL and education attainment 4 years after high school from adjusted multinomial logistic regression models. Compared with DDL, no-DDL was associated with a higher likelihood of completing a bachelor's degree or graduate degree (AOR = 2.55, 95% CI = 1.52, 4.28) and some college/technical school or an associate's degree (AOR = 1.89, 95% CI = 1.21, 2.97) compared with a high school diploma or less than a high school diploma or a GED.

Table 4 shows the associations between DDL and employment 4 years after high school from adjusted multinomial logistic regression models. Compared with DDL, no-DDL was associated with a higher likelihood of working ≥ 30 hours/week (AOR = 7.53, 95% CI = 1.40, 40.48) and working < 30 hours/week (AOR = 1.47, 95% CI = 1.04, 2.09) compared with not working.

Discussion

To the best of our knowledge, this study is one of the first studies to assess the association of teen DDL with their later

Table 3

Adjusted multinomial logistic regressions of education attainment at W7 on delaying driving licensure

	4-year college+			Tech/community college		
	AOR	95% CI	p	AOR	95% CI	p
DDL (≥ 1 yr.)	Ref.	–	–	Ref.	–	–
No-DDL	2.55	1.52, 4.28	<.001	1.89	1.21, 2.97	.005

Note: Delay is the reference group; Logit models used high school or less as the reference category.

AOR = adjusted odds ratio; CI = confidence interval; DDL = delaying driving licensure.

^aadjusting for sex, race/ethnicity, family affluence, parental education, urbanicity, and overall physical health at W1.

Table 4

Adjusted multinomial logistic regressions of employment indicated by working hours at W7 on delaying driving licensure

	≥30 hours			<30 hours		
	AOR	95% CI	p	AOR	95% CI	p
DDL (≥1 yr.)	Ref.	–	–	Ref.	–	–
No-DDL	7.53	1.40, 40.48	.019	1.47	1.04, 2.09	.031

Note: Delay is the reference group; Logit models used none working hours per week as the reference category.

AOR = adjusted odds ratio; CI = confidence interval; DDL = delaying driving licensure.

^aadjusting for sex, race/ethnicity, family affluence, parental education, urbanicity, and overall physical health at W1.

health, education, and employment when transitioning to early adulthood. Our study shows that adolescents who obtained their driver's license within a year of legal eligibility were more likely to report better health status, higher education attainment, and more employment than those who delayed driving licensure.

Our findings are consistent with previous studies indicating that the increase in independence and opportunities provided by personal transportation may enhance short- and long-term well-being/health, education, and employment [2,3]. More specifically, obtaining independent driving licensure might increase emerging adults' access to health care. For example, independent transportation could help those who live in geographic regions further away from health care resources overcome large distances between residences and services and more freely travel to health care professional and care centers when needed [2]. Apart from having access to health care resources, the greater independence due to licensure may offer a variety of important opportunities such as supplemental education, extracurricular activities, and employment seeking if those opportunities are at a distance from their homes [3]. Thus, the potential benefits of such opportunities facilitated by independent transportation could boost teens' future health, education, and employment when they transition to the broader developmental, social, and environmental context of emerging adulthood [10,11].

The potential advantages of earlier licensure (within state-specific legal eligibility) must be balanced against the relatively earlier and greater driving exposure and related risk [27]. It is well understood that with greater exposure to driving, particularly at younger ages, there is heightened risk for crash [27]. Despite the added risk due to exposure, for many teens and their families, the licensure can increase independence, and opportunities for enhanced short- and longer-term well-being/health, education, and employment are enough to outweigh the associated risks of teen licensure and risky driving [2,3,10,11]. Our findings highlight short-, intermediate-, and potentially long-term benefits of independent driving licensure (licensure typically gained through participating in a state GDL program) at important transitional periods for the development of youth.

The rate of new driver licensure among high school seniors has continued to decrease [12]. Previous studies suggest that this reduction trend in teen driving licensure may be linked to GDL policy restrictions (e.g., permit holding periods, initial legal eligibility time of licensure, night time driving, number of passengers in the vehicle) [13,14] and economics of driving [28]. The upward trend of DDL among adolescents is of note because young drivers who delay licensure may miss out on GDL driving safety instructions and restrictions that have been shown to

enhance novice driver safety. Currently, most state GDL policies only apply to young drivers aged ≤ 18 y/o [29]. Thus, those who miss out on GDL owing to DDL beyond age 18 y/o may potentially heighten their risk of crash when and if they pursue licensure at a later age [28]. In line with previous evidence, young drivers who DDL may compromise later beneficial opportunities related to health, education, and employment [18,19].

The findings from our study imply that the opportunities for young drivers to flourish in emerging adulthood might be enhanced if teens pursued licensure soon after they reach the age of state legal eligibility. Of course, one needs to take into consideration the individual variability that exists among teens when it comes to being developmentally ready to drive at the earliest state legal age. Nevertheless, not delaying licensure could assure teen exposure to the many driver safety benefits of GDL, although, admittedly, it would also increase driving exposure and related risk, but this would unlikely be beyond what is already present when adhering to state-level licensure policies for teen drivers.

There are limitations to our study. First, it would have been optimal to know the exact date (i.e., month/date/year) when participants received their driving license so that an exact DDL measure could have been used in our analysis. Given that the exact licensure dates were not available in the NEXT study, we calculated and approximated the DDL variable as any delay in licensure past the earliest assessment when a participant was eligible for licensure with consideration of their state's legal requirements. Second, measures of health, education, and employment were based on annual self-reported questions, and this inherently introduces potential for social desirability and recall bias. In addition, employment was measured with self-reported paid and/or unpaid jobs, and this does not differentiate employment from volunteer activities. This could overestimate the time in paid employment. However, many volunteer activities represent valuable educational opportunities. Third, the NEXT survey was not designed to explore crash risk among adolescents and emerging adult participants. As such, we are not able to empirically test any speculated association between DDL-related heightened crash risk and its relationship to later health, education, and employment. There remains a need to empirically explore and examine this link.

Despite these limitations, our research has notable strengths. Our study was based on a longitudinal design that allows us to examine the long-term association of teen DDL with their later health, education, and employment 4 years after high school. In addition, we used a nationally representative sample that allows for sound national generalizability to the US emerging adult populations.

In conclusion, adolescents with no delay in driving licensure had a higher likelihood of self-reported better health status, higher education attainment, and more employment as emerging adults. This suggests that timely, legally eligible teen driving licensure could provide advantages for health, education, and employment during early adulthood. The findings of this study can provide families and injury prevention programs with important additional facets of information when considering the timing and pursuit of licensure for adolescents and teen driver safety education.

Supplementary Data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jadohealth.2022.01.128>.

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